

Technical Session on

Correction/Exception Manual

For MASONRY STRUCTURE

for houses that have built under the
HOUSING RECONSTRUCTION PROGRAMME

Cover Page of
Manual



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Meeting Hall, DUDBC

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Objectives of Presentation

▪ Main Objectives :

- To provide overview on Correction/Exceptional manual to Partner Originations so that they can help in detail training in later days.

- ✓ Inform about Number of Trained Engineer's (MToT 76) .

- Brief about updated inspection forms

Content of Presentation

■ Overall Contents of the Presentation :

- ✓ High Lights Few “TERMINOLOGY” and its Definition
- ✓ Brief on Contents of Manual/each Chapter
- ✓ Brief on Layout of Chapter
- ✓ Present Few Examples from Manual for Clarity

Contents in Manual

- **Chapters in Manual Mainly Covers:**

- Background of Manual
- Scope and Limitation of Manual
- Part - A : Exception/ Correction
- Part – B : Mitigation Measures
- Annexes - 1 & 2

Background of Manual

- Grant is disbursed based on construction compliance with MRs, NBCs, etc.
- Many houses have been found to be non-compliant during inspection.
- Non-compliant Buildings are vulnerable to future Earthquakes.
- This manual helps site engineers in providing advice and guidance to households for correction and to be eligible for tranches.

Scope and Limitation

- **Applicability** : This Manual is applicable only for the **residential houses** that have been constructed after Gorkha earthquake 2015 under the GoN reconstruction programme
- The manual intends to achieve the minimum acceptable structural safety envisioned in NBC 105 : 1994.
- Designs are **ready-to-use designs** for all structural components, some are **set as advisory** measures.

Scope and Limitation...

- **Limitation** : For Load Bearing Masonry Buildings (new or under construction) **ONLY**.



- ✓ **Note** : If the construction up to plinth level is as per requirement, application can be recommended for second tranches. Correction or mitigation works at super structure should be made before third tranche.

Part – A : Exception/correction

- **Exception** : is the cases of the buildings that do not comply with MRs but are structural safe as per NBC 105.
- **Correction** : is corrective measures required to make newly constructed or under construction buildings compliant with the seismic resistant standard as per NBC 105

Part – A : Exception/correction...

Chapter lay out

- ✓ What is in the Minimum Requirements?
- ✓ Problem Statements.
- ✓ Common Defects Found in Field.
- ✓ Exceptions : Presents Engineering judgement and Residual Safety Evaluation.
- ✓ Solutions : Covers alternative corrective measures

Part – A : Exception/correction...

- Topics Heading it covers

- Site Selection
- Shape of size of Building
- Materials
- Foundation
- Vertical Member

Part – A : Exception/correction...

- Topics Heading it covers...

- Plinth Beams
- Walls
- Doors and Windows openings
- Horizontal Bands
- Roof

Part – A : Exception/correction...

- Some Examples

- ✓ Site Selection

- ✓ Shape and Size of Buildings

Site Selection

- Site selection shall be done so as to minimize risks in relation to natural hazards. No buildings shall be constructed in hazardous areas:

- Steep slope area

- ✓ Constructed retaining wall



Shape and Size of Building

- Addition of X-walls, Buttress Walls etc.

Correction

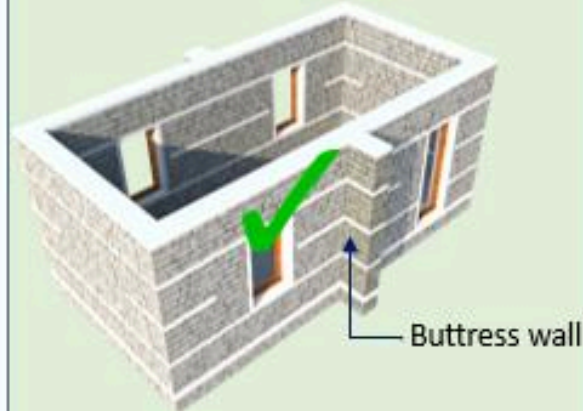
Span and size of room is more than in MRS



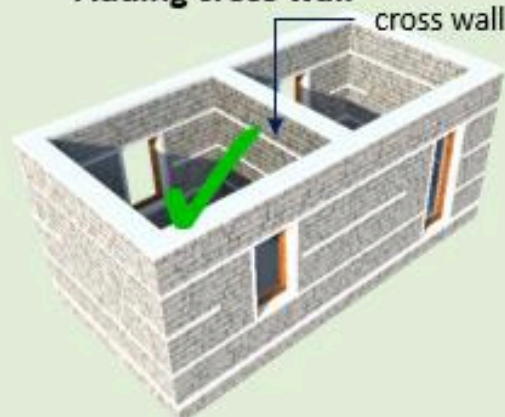
P35 ~ and P36 ~

*See Mitigation Measures 2 and 3, for advice on how to construct buttress and cross wall

Adding buttress wall

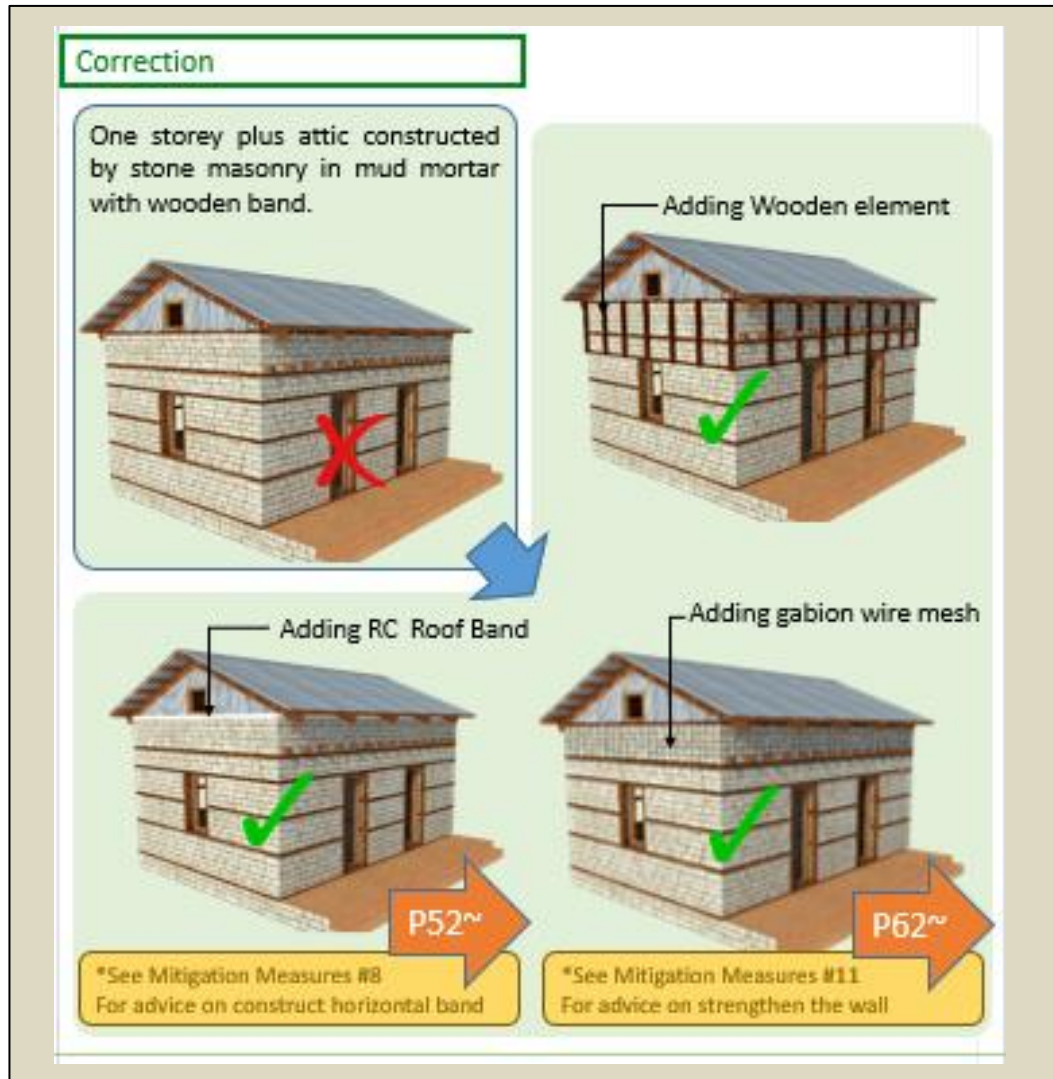


Adding cross wall



Shape and Size of Building...

- Strengthening using Wood, GI wires



Part – B : Mitigation Measures

- It Covers :

- Step by step procedures for implementing different corrective measures
- Material Specification
- Construction Sequence and Procedures
- Photographic Images for Clarity

Part – B : Mitigation Measures...

- Headings:

- Retaining wall
- Buttress
- Cross wall
- Vertical & Horizontal members (using RC/GI wires/ Wood)

Part – B : Mitigation Measures...

■ Heading : ...

- Retaining wall
- Buttress
- Cross wall
- Vertical & Horizontal members (using RC/GI wires/ Wood)
- Strengthen Openings
- Strengthen Wall by Jacketing

Annex – 1

- **Annex** : 1 gives a solution to the building which missed all the earthquake resistant elements
- Solution are cost effective and has highly constructability options (six options)
- Materials used are wood, Iron bars, GI Wires, Precast concrete sections

Annex – 2

- **Annex – 2** : gives a solution to mixed/hybrid building typology such as : ground floor BMC/SMC and light or timber structure above.
- **Option 1 to 4** : use of Wooden/Bamboo Section (inside and outside)
- Option 5: Use of Iron Section
- Option 6 : Use of Iron Section with Prefab Panel

Inspection Forms

- First and Second Level Forms

✓ **New form for Confined Masonry Buildings**

Thank you

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